RMIT UNIVERSITY

CSIRO

RMIT Quantum Summer School 2026

Project

Deutsch–Jozsa's Algorithm

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The Deutsch–Jozsa's Problem

Given a **2-qubit** Oracle ("black box") function

We don't know exactly what is inside, but we know the promise that it is either :

- **constant** (i.e., always returns 0 or 1 whenever it's queried)

or

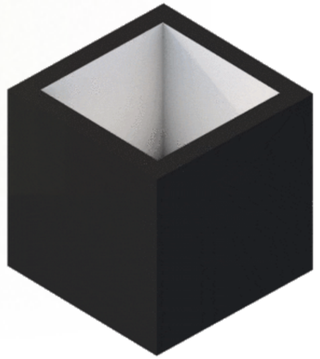
- **balanced** (i.e., 50% returns 0 and 50% returns 1).

→ Determine whether that oracle is constant or balanced

With classical solution, how many query do we need?

- ***The best case: 2 queries***
- ***The worst case: 3 queries***

How about Quantum solution? ***→ 1 query!***



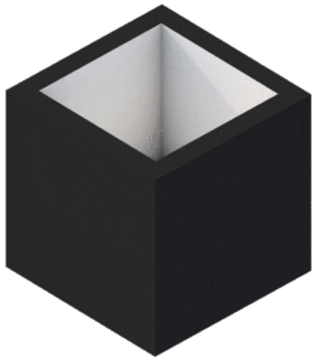
Oracle

- **Input**: 00, 01, 10, 11
- **Output**: 0 or 1

Project: Deutsch-Jozsa's Algorithm

Project Overview

The Deutsch-Jozsa algorithm (1992) was the first quantum algorithm to prove exponential speedup over classical computation. In this project, you will **implement** and **analyse** the **Deutsch-Jozsa algorithm** for **2-qubit black-box** functions.



Oracle

- **Input**: n-bit
- **Output**: 0 or 1

Deliverables

1. Multiple **oracle** functions demonstrating **constant** and **balanced** cases
2. Working implementation of the **2-qubit Deutsch-Jozsa algorithm**
3. Comparison of quantum query complexity vs. classical approaches

Advanced *(optional)*

1. Working implementation of the **n-qubit Deutsch-Jozsa algorithm**
2. Analysis of why the DJ algorithm works through the phase kickback mechanism.
3. Analysis of **circuit depth** and **gate complexity as n scales**
4. Testing and validation on ideal and noisy quantum simulators